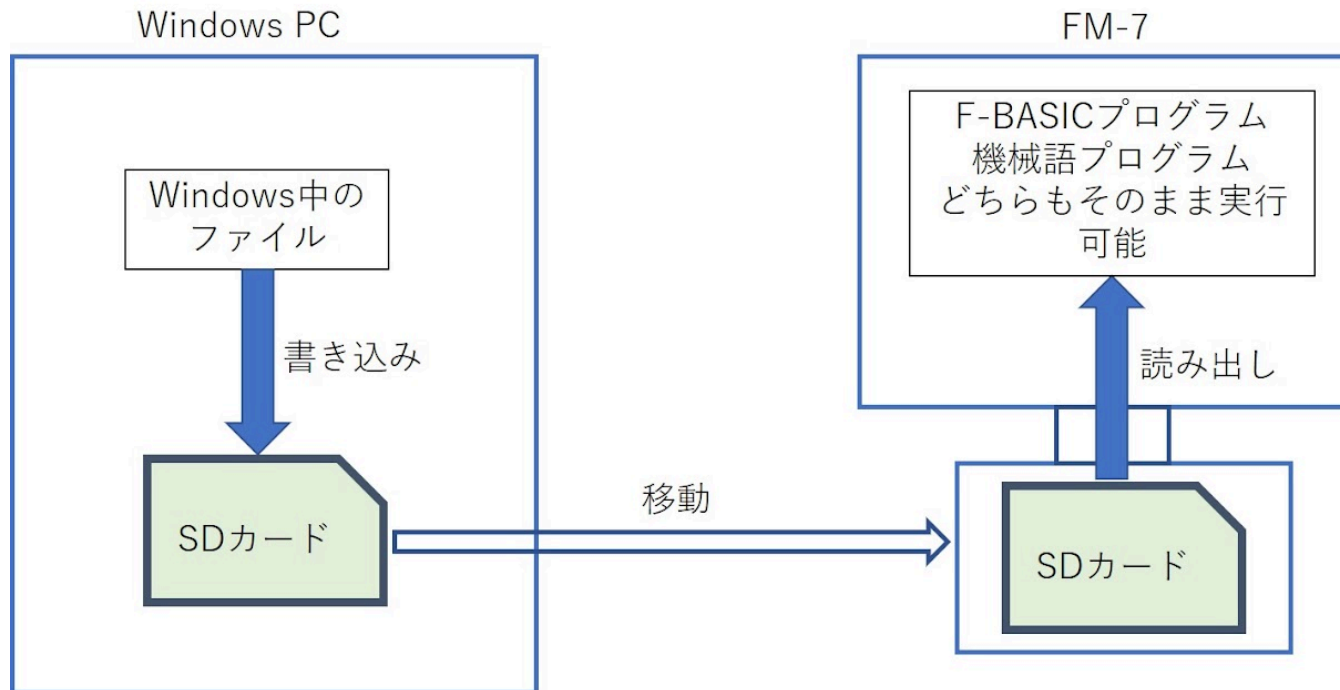


An attempt to connect RTC and SD card to FM-7 (SD card version, part 2)

 flexonsbd.blogspot.com/2023/03/fm-7rtcscsd_25.html

Saturday, March 25, 2023



An attempt to connect an RTC and SD card to the FM-7 (SD card version, part 2)

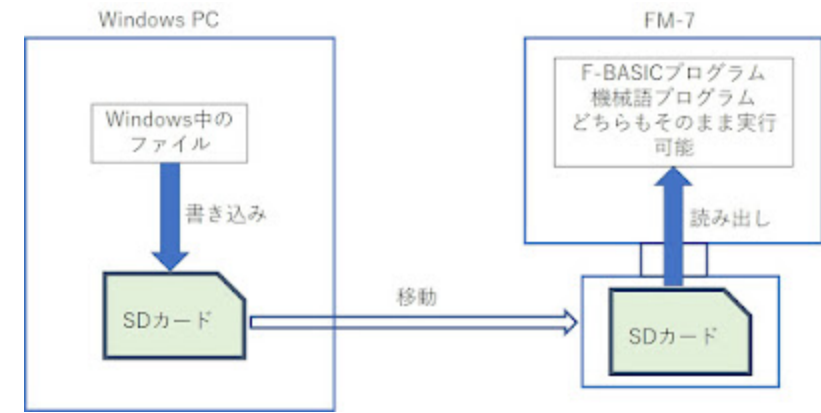
I was able to read and execute BASIC program files and machine code files from the SD card.

Writing to the SD card is not yet possible, but reading from it is now feasible.

Ideally, I'd like to be able to handle SD cards in the same way as floppy disks using F-BASIC commands (FILES, LOAD, LOADM), but since each sector is only 512 bytes in size, it seems difficult for me to achieve this.

For now, assuming its use as a tool to transfer files from a Windows PC to an FM-7 using an SD card, I've created a dedicated program that launches each time to read the SD card, rather than being a resident program.

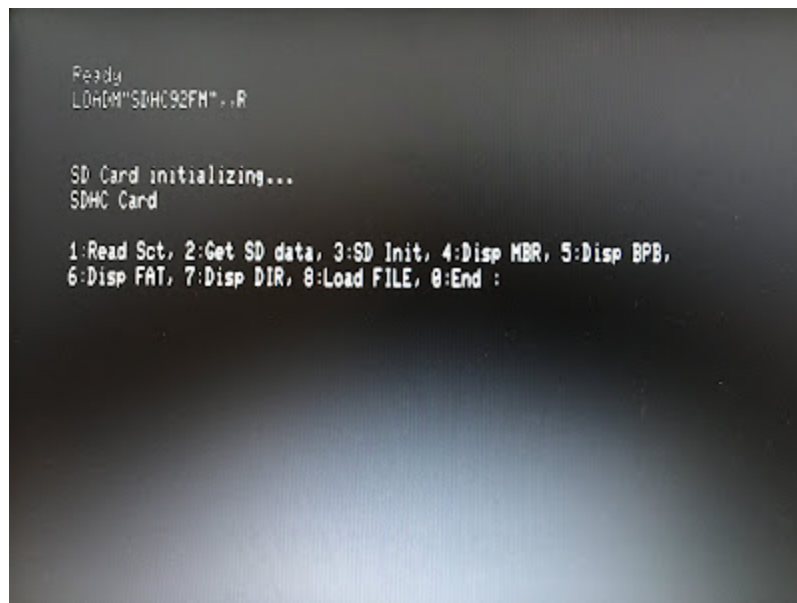
The usage is illustrated in the diagram below.



Usage image

With the standard FT245 high-speed communication card, the entire disk image is copied, but with this SD card, you can select specific files to transfer to the FM-7.

This is the program's startup screen.



Startup screen

Menu items are

1. Read the specified sector
2. Data acquisition and display from the SD card
3. Initialize the SD card
4. Displaying the contents of the MBR (Master Boot Record)

5. Displaying the contents of the BPB (Boot Parameter Block)
 6. Displaying the contents of the FAT (File Allocation Table)
 7. Displaying the contents of the DIR (Directory)
 8. When loading the specified file
- , what is the difference from last time?
- The writing to the sector was omitted.
 - Changed the directory display format and enabled the display of all pages.
 - Instead of displaying the file on the screen, the data was loaded into memory and integrated with F-BASIC.

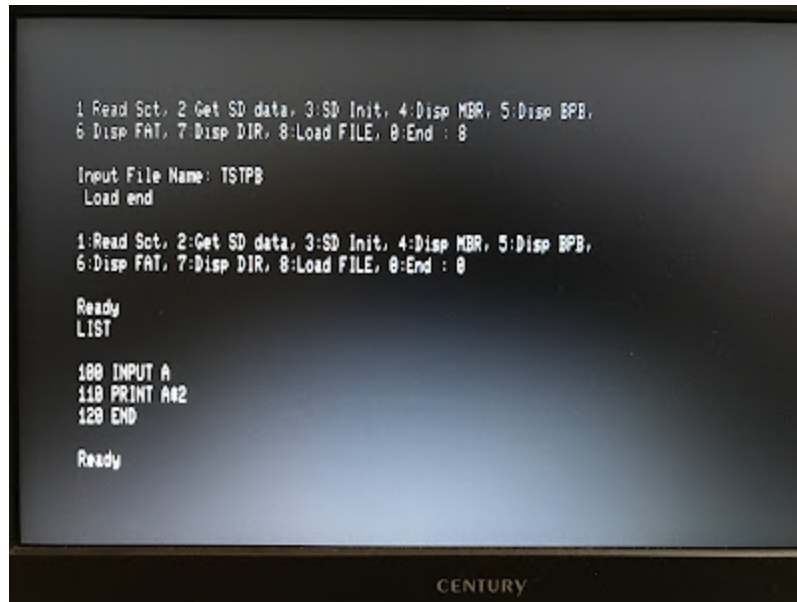
The scene is shown below.

Instead of using a number like last time, you select files by entering their filenames.



Directory display and file loading

The BASIC program loaded into memory can be displayed and executed normally, as shown in the image below.



List the loaded BASIC files.

Although not shown here, machine code files are also loaded to the specified address.

To summarize the features we have implemented so far:

1. It supports SD cards and SDHC cards.
2. Files written to the SD card can be read normally on Windows.
3. It can handle F-BASIC program files and machine code files.

The restrictions are

1. This software is designed for use with the FM-7, and therefore does not support large files exceeding 32KB.
2. File names must not include extensions. (Extensions are not checked.)
3. Lowercase letters cannot be used in filenames. (They will be converted to uppercase.)
4. F-BASIC program files can only be handled if they have been converted to intermediate language; ASCII saved files cannot be used. (Because they are written directly to the text area.)
5. The machine language file must be in the F-BASIC disk save format.

Although there are many limitations, I think it could be used as a tool to transfer BASIC program files and machine code files saved on Windows to the FM-7 via an SD card.

Further things I want to achieve

The remaining challenge is, of course, to implement the functionality to write programs and other files to an SD card using the FM-7 and then read them on Windows. Once this is achieved, the project will be complete.

However, this involves rewriting the FAT, which brings in the FMS (File Management System), and since the program size is already close to 3KB, if it gets any larger, where to place it will become an issue.

This concludes my report on successfully reading files from an SD card using 6809 assembler. Note that only three routines from the FM-7 itself are used: INCHA, OUTCHA, and CSRON.

The program I created is shown below.

```

1      0001      *
2      0002      *   SDHC9.TXT
3      0003      *
4      0004      *   SD Read/Write for FM-7
5      0005      *
6      0006      *       use RTC & SD Card
7      0007      *
8      0008      *       6821(PIA) Port A <--> RTC (I2C Mode)
9      0009      *
10     0010      *           SBC           RTC           Ext I0
11     0011      *   CA2  2  -->           -->  HV4 LV4 -->  I01
12     0012      *   PA0  3  -->  SCL  -->  HV1 LV1 -->  I04
13     0013      *   PA1  4  <-- --="  0014="  10="  hv2="  io3="  lv2="
14     pa7=" ">  SDA  -->  HV3 LV3 -->  I02
15     0015      *   GND      --           --           --  GND
16     0016      *           --           --           --  3V3
17     0017      *
18     0018      *
19     0019      *       6821(PIA) Port B <--> SD (SPI Mode)
20     0020      *
twenty 0021      *           SBC           SD
one     0022      *   CB2 12  -->  HV4 LV4 -->  SCK
twenty 0023      *   PB0 13  -->  HV1 LV1 -->  CS
two     0024      *   PB1 14  <-- --="  0025="  20="  hv2="  lv2="
twenty  miso=" " pb7=" "> HV3 LV3 -->  MOSI
three   0026      *   GND 31  --           --  GND
twenty  0027      *   VCC 33  --           --  VCC
four    0028      *
twenty  0029      *       cf. all data line use level converter
five    0030      *
26      0031      *       PB  output mode
27      0032      *       CB2 1bit output
28      0033      *
29      0034      *       READ/WRITE BUFFER  $6D00 - 6EFF
30      0035      *
31      0036      *       Load data to F-BASIC '2023.03.23
32      0037      *       Add command           '2023.03.14
33      0038      *       Change use Port B      '2023.03.07

```

34	0039	*	Original version				'2022.06.11	
35	0040	*	Copyright by Y.Yamamoto					
36	0041	*						
37	0042	6000	STTADR	EQU	\$6000			
38	0043							
39	0044	FBB8	INCHA	EQU	\$FBB8			
40	0045	FBBB	OUTCHA	EQU	\$FBBB			
41	0046	FBF7	CSR0N	EQU	\$FBF7			
42	0047							
43	0048	6D00	RWBUF	EQU	STTADR+\$0D00			
44	0049							
45	0050	FD9E	PBDAT	EQU	\$FD9E			
46	0051	FD9F	PBCTL	EQU	PBDAT+1			
47	0052	0033	BASADRP	EQU	\$33	pointer of adrs for F-		
48	BASIC							
49	0053	2000	TXTADR	EQU	\$2000	for TEXT		
50	0054							
51	0055	6000	ORG	STTADR				
52	0056							
53	0057	6000	20	58	START	BRA	MAIN	
54	0058	6002	01	VN	FCB	1		
55	0059	6003	00	SCTN0	FCB	0	top adrs of sct	
56	0060	6004	00	SCTN1	FCB	0		
57	0061	6005	00	SCTN2	FCB	0		
58	0062	6006	00	SCTN3	FCB	0		
59	0063	6007	00	RESDAT	FCB	0	response data	
60	0064	6008	00	SDHC	FCB	0	sd:0,sdhc:1	
61	0065							
62	0066	6009	0000	SCTBPB	FDB	00	sct no of BPB(4bytes)	
63	0067	600B	0000		FDB	00		
64	0068	600D	0000	SCTFAT	FDB	00	sct no of	
65	FAT#1(4bytes)							
66	0069	600F	0000		FDB	00		
67	0070	6011	0000	SCTDIR	FDB	00	sct no of DIR(4bytes)	
68	0071	6013	0000		FDB	00		
69	0072	6015	0000	SCTDSV	FDB	00	save sct no of DIR	
70	0073	6017	0000		FDB	00		
71	0074	6019	0000	SCTCLT	FDB	00	sct no of top	
72	CLUSTER(4bytes)							

```

73      0075  601B  0000          FDB      00          if SD:#2, SDHC:#3
74      0076  601D  00      SCTPCL  FCB      0          sct per cluster
75      0077  601E  FF      DIRENT  RMB      32
76      0078  603E  FF      WMDY    RMB      3
77      0079  6041  FF      WFSIZ   RMB      4
78      0080  6045  FF      FILSIZ  RMB      2
79      0081  6047  0000      CLSTH   FDB      00
80      0082  6049  0000      CLSTL   FDB      00
81      0083  604B  00      CNT      FCB      0
82      0084  604C  00      CNTSV   FCB      0
83      0085  604D  00      MCFLG   FCB      0          1:machine code, 0:else
84      0086  604E  00      DPGFLG  FCB      0          1:1st page of DIR,
85      0:else
86      0087  604F  FF      FNAME   RMB      9
87      0088  6058  0000      WK1     FDB      00
88      0089
89      0090  605A          MAIN     EQU      *
90      0091          *=====reset pia=====
91      0092          *portB:output
92      0093  605A  86  30          LDA      #$30
93      0094  605C  B7  FD9F          STA      PBCTL          select ddra, cb2:0
94      0095  605F  86  FD          LDA      #$FD          pb0,7:out,pb1:in
95      0096  6061  B7  FD9E          STA      PBDAT          set output mode
96      0097  6064  86  34          LDA      #$34
97      0098  6066  B7  FD9F          STA      PBCTL          select
98      peripheral,cb2:0
99      0099
100     0100          *for pia exist check
101     0101  6069  86  55          LDA      #$55          test data
102     0102  606B  B7  FD9E          STA      PBDAT
103     0103  606E  B6  FD9E          LDA      PBDAT
104     0104  6071  84  FD          ANDA      #$FD          pb1=0
105     0105  6073  81  55          CMPA      #$55
106     0106  6075  27  07          BEQ      INIT
107     0107  6077  8E  6A30          LDX      #MERR
108     0108  607A  17  075F          LBSR      PSTRNG
109     0109  607D  39          RTS          return to BASIC
110     0110
111     0111  607E          INIT      EQU      *

```

```

112 0112 607E 8D 59          BSR      SDINIT
113 0113
114 0114          *****
115 0115          * MAIN MENU
116 0116          *****
117 0117 6080          MENU      EQU      *
118 0118 6080 8E 6976          LDX      #MSMENU1
119 0119 6083 17 0756          LBSR      PSTRNG
120 0120 6086 8E 69B6          LDX      #MSMENU2
121 0121 6089 17 0750          LBSR      PSTRNG
122 0122 608C 17 9B68          LBSR      CSRON      cursor on
one 0123
two 0124 608F BD 690B          JSR      INOUTC
three 0125 6092 80 30          SUBA      #$30
124 0126 6094 26 04          BNE      MEN1      0?
125 0127 6096 17 068E          LBSR      CRLF
126 0128 6099 39          RTS      return to BASIC
127 0129
128 0130 609A 4A      MEN1     DECA      1?
129 0131 609B 26 05          BNE      MEN2
130 0132 609D 17 0279          LBSR      SDRD
131 0133 60A0 20 DE          BRA      MENU
132 0134 60A2 4A      MEN2     DECA      2?
133 0135 60A3 26 05          BNE      MEN3
134 0136 60A5 17 01E3          LBSR      GETSD
135 0137 60A8 20 D6          BRA      MENU
136 0138 60AA 4A      MEN3     DECA      3?
137 0139 60AB 26 04          BNE      MEN4
138 0140 60AD 8D 2A          BSR      SDINIT
139 0141 60AF 20 CF          BRA      MENU
140 0142 60B1 4A      MEN4     DECA      4?
141 0143 60B2 26 05          BNE      MEN5
142 0144 60B4 17 0275          LBSR      DSPMBR
143 0145 60B7 20 C7          BRA      MENU
144 0146 60B9 4A      MEN5     DECA      5?
145 0147 60BA 26 05          BNE      MEN6
146 0148 60BC 17 02C7          LBSR      DSPBPB
147 0149 60BF 20 BF          BRA      MENU
148 0150 60C1 4A      MEN6     DECA      6?

```

```

149 0151 60C2 26 05          BNE      MEN7
150 0152 60C4 17 02D1          LBSR      DSPFAT
151 0153 60C7 20 B7          BRA      MENU
152 0154 60C9 4A  MEN7  DECA          7?
153 0155 60CA 26 05          BNE      MEN8
154 0156 60CC 17 02DB          LBSR      DSPDIR
155 0157 60CF 20 AF          BRA      MENU
156 0158 60D1 4A  MEN8  DECA          8?
157 0159 60D2 26 AC          BNE      MENU
158 0160 60D4 17 0398          LBSR      LDFILE
159 0161 60D7 20 A7          BRA      MENU
160 0162          *
161 0163          *****
162 0164          * SD Initialize routine
163 0165          *****
164 0166 60D9          SDINIT  EQU      *
165 0167          *----- start -----
166 0168 60D9 8E 6A14          LDX      #MSINIT
167 0169 60DC 17 06FD          LBSR      PSTRNG
168 0170          * cs:1,di:1,sck>74ms
169 0171 60DF 86 FF          LDA      #$FF
170 mosi:b7=H,miso:b1=H(for PIA),cs:b0=H
171 0172 60E1 B7 FD9E          STA      PBDAT
172 0173 60E4 C6 50          LDB      #80      80times
173 0174 60E6 17 0603  ISTT  LBSR      SETCKL
174 0175 60E9 17 0609          LBSR      SETCKH
175 0176 60EC 5A          DECB
176 0177 60ED 26 F7          BNE      ISTT
177 0178          *
178 0179          *----- cmd 0 -----
179 0180 60EF          SCMD0  EQU      *
180 0181          * LDX #MCMD0
181 0182          * LBSR PSTRNG
182 0183
183 0184 60EF 17 05FA          LBSR      SETCKL
184 0185 60F2 C6 06          LDB      #6
185 0186 60F4 8E 6912          LDX      #CMD0
186 0187 60F7 A6 80 SC01  LDA      ,X+
187 0188 60F9 17 057B          LBSR      SNDDAT

```

188	0189	60FC	5A		DECB	
189	0190	60FD	26	F8	BNE	SC01
190	0191				*get response	
191	0192	60FF	5F		CLRB	
192	0193	6100		GR00		
193	0194	6100	5A		DECB	
194	0195	6101	1027	020D	LBEQ	SDERR
195	0196	6105	17	058B	LBSR	GETDAT
196	0197	6108	B6	6007	LDA	RESDAT
197	0198				* LBSR DSPBIN disp response	
198	0199	610B	81	01	CMPA	#\$01
199	0200	610D	26	F1	BNE	GR00
200	0201				* LBSR CRLF	
201	0202				*	
202	0203				*----- cmd 8 -----	
203	0204	610F		SCMD8	EQU	*
204	0205				* LDX #MCMD8	
205	0206				* LBSR PSTRNG	
206	0207					
207	0208	610F	17	05DA	LBSR	SETCKL
208	0209	6112	86	FF	LDA	#\$FF
						dummy cmd
209	0210	6114	17	0560	LBSR	SNDDAT
210	0211	6117	C6	06	LDB	#6
211	0212	6119	8E	692A	LDX	#CMD8
212	0213	611C	A6	80 SC80	LDA	,X+
213	0214	611E	17	0556	LBSR	SNDDAT
214	0215	6121	5A		DECB	
215	0216	6122	26	F8	BNE	SC80
216	0217				*get response	
217	0218	6124		GR80		
218	0219	6124	17	056C	LBSR	GETDAT
219	0220	6127	B6	6007	LDA	RESDAT
220	0221					
221	0222	612A	17	0566	LBSR	GETDAT
222	0223	612D	B6	6007	LDA	RESDAT
223	0224				* LBSR DSPBIN disp response	
224	0225	6130	81	01	CMPA	#\$01
225	0226	6132	26	F0	BNE	GR80
226	0227	6134	C6	04	LDB	#4

```

227 0228 6136 17 055A GR81 LBSR GETDAT
228 0229 6139 B6 6007 LDA RESDAT
229 0230 * LBSR DSPBIN disp response
230 0231 613C 5A DECB
231 0232 613D 26 F7 BNE GR81
232 0233 * LBSR CRLF
233 0234 *
234 0235 *----- cmd 58 -----
235 0236 613F SCMD58 EQU *
236 0237 * LDX #MCMD58
237 0238 * LBSR PSTRNG
238 0239
239 0240 613F 17 05AA LBSR SETCKL
240 0241 6142 86 FF LDA #$FF dummy cmd
241 0242 6144 17 0530 LBSR SNDDAT
242 0243 6147 C6 06 LDB #6
243 0244 6149 8E 6930 LDX #CMD58
244 0245 614C A6 80 SC581 LDA ,X+
245 0246 614E 17 0526 LBSR SNDDAT
246 0247 6151 5A DECB
247 0248 6152 26 F8 BNE SC581
248 0249 *get response
249 0250 6154 GR580
250 0251 6154 17 053C LBSR GETDAT
251 0252 6157 B6 6007 LDA RESDAT
252 0253 * LBSR DSPBIN disp response
253 0254 615A 81 01 CMPA #$01
254 0255 615C 26 F6 BNE GR580
255 0256 615E C6 04 GR581 LDB #4
256 0257 6160 17 0530 GR582 LBSR GETDAT
257 0258 6163 B6 6007 LDA RESDAT
258 0259 * LBSR DSPBIN disp response
259 0260 6166 5A DECB
260 0261 6167 26 F7 BNE GR582
261 0262 6169 RPT41 EQU *
262 0263 *----- cmd 55 -----
263 0264 6169 SCMD55 EQU *
264 0265 * LDX #MCMD55
265 0266 * LBSR PSTRNG

```

266	0267							
267	0268	6169	17	0580	LBSR	SETCKL		
268	0269	616C	86	FF	LDA	#\$FF	dummy cmd	
269	0270	616E	17	0506	LBSR	SNDDAT		
270	0271	6171	C6	06	LDB	#6		
271	0272	6173	8E	6936	LDX	#CMD55		
272	0273	6176	A6	80 SC551	LDA	,X+		
273	0274	6178	17	04FC	LBSR	SNDDAT		
274	0275	617B	5A		DECB			
275	0276	617C	26	F8	BNE	SC551		
276	0277				*get response			
277	0278				* CLRB			
278	0279	617E			GR550			
279	0280				* DECB			
280	0281				* BEQ SDERR			
281	0282	617E	17	0512	LBSR	GETDAT		
282	0283	6181	B6	6007	LDA	RESDAT		
283	0284				* LBSR DSPBIN disp response			
284	0285	6184	81	01	CMPA	#\$01		
285	0286	6186	26	F6	BNE	GR550		
286	0287							
287	0288				* LBSR CRLF			
288	0289				*----- acmd 41 -----			
289	0290	6188			ASCMD41 EQU	*		
290	0291				* LDX #MCMD41			
291	0292				* LBSR PSTRNG			
292	0293							
293	0294	6188	17	0561	LBSR	SETCKL		
294	0295	618B	86	FF	LDA	#\$FF	dummy cmd	
295	0296	618D	17	04E7	LBSR	SNDDAT		
296	0297	6190	C6	06	LDB	#6		
297	0298	6192	8E	693C	LDX	#ACMD41		
298	0299	6195	A6	80 ASC411	LDA	,X+		
299	0300	6197	17	04DD	LBSR	SNDDAT		
300	0301	619A	5A		DECB			
301	0302	619B	26	F8	BNE	ASC411		
302	0303				*get response			
303	0304	619D			GR410			
304	0305	619D	17	04F3	LBSR	GETDAT		

```

305 0306 61A0 B6 6007          LDA    RESDAT
306 0307          * LBSR DSPBIN disp response
307 0308 61A3 17 04ED          LBSR    GETDAT
308 0309 61A6 B6 6007          LDA    RESDAT
309 0310          * LBSR DSPBIN disp response
310 0311 61A9 81 00          CMPA    #$00
311 0312 61AB 26 BC          BNE     RPT41
312 0313          * LBSR CRLF
313 0314
314 0315          *----- second cmd 58 -----
315 0316 61AD          SCMD258 EQU    *
316 0317          * LDX #MCMD58
317 0318          * LBSR PSTRNG
318 0319
319 0320 61AD 17 053C          LBSR    SETCKL
320 0321 61B0 86 FF          LDA     #$FF      dummy cmd
321 0322 61B2 17 04C2          LBSR    SNDDAT
322 0323 61B5 C6 06          LDB     #6
323 0324 61B7 8E 6930          LDX     #CMD58
324 0325 61BA A6 80 SC2581 LDA     ,X+
325 0326 61BC 17 04B8          LBSR    SNDDAT
326 0327 61BF 5A          DECB
327 0328 61C0 26 F8          BNE     SC2581
328 0329          *get response
329 0330 61C2          GR2580
330 0331 61C2 17 04CE          LBSR    GETDAT
331 0332 61C5 B6 6007          LDA    RESDAT
332 0333          * LBSR DSPBIN disp response
333 0334 61C8 81 00          CMPA    #$00
334 0335 61CA 26 F6          BNE     GR2580
335 0336 61CC 17 04C4          LBSR    GETDAT
336 0337 61CF B6 6007          LDA    RESDAT
337 0338          * LBSR DSPBIN disp response
338 0339 61D2 48          LSLA
339 0340 61D3 48          LSLA
340 0341 61D4 24 03          BCC     GR2581
341 0342 61D6 7C 6008          INC     SDHC      if sdhc then 1
342 0343 61D9 C6 03 GR2581 LDB     #3
343 0344 61DB 17 04B5 GR2582 LBSR    GETDAT

```

```

344 0345 61DE B6 6007          LDA    RESDAT
345 0346          * LBSR DSPBIN disp response
346 0347 61E1 5A              DECB
347 0348 61E2 26 F7          BNE     GR2582
348 0349
349 0350 61E4 8E 69FA          LDX     #MSSD0
350 0351 61E7 B6 6008          LDA     SDHC
351 0352 61EA 27 03            BEQ     SD1
352 0353 61EC 8E 6A06          LDX     #MSSD1
353 0354 61EF 17 05EA SD1      LBSR    PSTRNG
354 0355          *-----
355 0356          *set sct no of BPB
356 0357 61F2 17 0155          LBSR    GETMBR
357 0358 61F5 8E 6ECA          LDX     #RWBUF+446+8+4
358 0359 61F8 108E 6009          LDY     #SCTBPB
359 0360 61FC C6 04            LDB     #4
360 0361 61FE A6 82 ILOP1      LDA     , -X
361 0362 6200 A7 A0            STA     ,Y+
362 0363 6202 5A              DECB
363 0364 6203 26 F9          BNE     ILOP1
364 0365
365 0366          *set sct no of FAT#1
366 0367 6205 17 0165          LBSR    GETBPB
367 0368 6208 8E 6009          LDX     #SCTBPB
368 0369 620B 108E 600D          LDY     #SCTFAT
369 0370 620F C6 04            LDB     #4
370 0371 6211 A6 80 ILOP2      LDA     ,X+
371 0372 6213 A7 A0            STA     ,Y+
372 0373 6215 5A              DECB
373 0374 6216 26 F9          BNE     ILOP2
374 0375
375 0376 6218 8E 600D          LDX     #SCTFAT
376 0377 621B 108E 6D0E          LDY     #RWBUF+$0E      for SD & SDHC
377 0378 621F 17 0589          LBSR    ADD42
378 0379
379 0380          *set sct no of DIR
380 0381 6222 8E 600D          LDX     #SCTFAT
381 0382 6225 108E 6011          LDY     #SCTDIR
382 0383 6229 C6 04            LDB     #4

```

```

383 0384 622B A6 80 ILOP3 LDA ,X+
384 0385 622D A7 A0 STA ,Y+
385 0386 622F 5A DECB
386 0387 6230 26 F9 BNE ILOP3
387 0388
388 0389 6232 8E 6D16 LDX #RWBUF+$16 for SD
389 0390 6235 7D 6008 TST SDHC
390 0391 6238 27 03 BEQ SKP1
391 0392 623A C6 0E LDB #$24-$16
392 0393 623C 3A ABX for SDHC
393 0394 623D 1F 12 SKP1 TFR X,Y
394 0395 623F 8E 6011 LDX #SCTDIR
395 0396 6242 17 0566 LBSR ADD42
396 0397 6245 17 0563 LBSR ADD42 skip for FAT#2
397 0398
398 0399 *save no of directory
399 0400 6248 8E 6011 LDX #SCTDIR
400 0401 624B 108E 6015 LDY #SCTDSV
401 0402 624F C6 04 LDB #4
402 0403 6251 A6 80 SCPY1 LDA ,X+
403 0404 6253 A7 A0 STA ,Y+
404 0405 6255 5A DECB
405 0406 6256 26 F9 BNE SCPY1
406 0407
407 0408 *set sct per cluster
408 0409 6258 8E 6D0D LDX #RWBUF+$0D
409 0410 625B A6 84 LDA ,X
410 0411 625D B7 601D STA SCTPCL
411 0412
412 0413 *set sct no of CLUSTER#2
413 0414 6260 8E 6011 LDX #SCTDIR
414 0415 6263 108E 6019 LDY #SCTCLT
415 0416 6267 C6 04 LDB #4
416 0417 6269 A6 80 ILOP4 LDA ,X+
417 0418 626B A7 A0 STA ,Y+
418 0419 626D 5A DECB
419 0420 626E 26 F9 BNE ILOP4
420 0421
421 0422 6270 108E 605A LDY #WK1+2

```

```

422 0423 6274 86 00          LDA    #$00
423 0424 6276 A7 A2          STA    , -Y
424 0425 6278 86 20          LDA    #$20          for SD
425 0426 627A 7D 6008        TST     SDHC
426 0427 627D 27 03          BEQ     SKP2
427 0428 627F B6 601D        LDA     SCTPCL        for SDHC
428 0429 6282 A7 A2 SKP2     STA     , -Y
429 0430 6284 8E 6019        LDX     #SCTCLT
430 0431 6287 17 0521        LBSR    ADD42
431 0432          *-----
432 0433 628A 39              RTS
433 0434
434 0435          *****
435 0436          * GET SD DATA ROUTINE
436 0437          * use CMD9
437 0438          *****
438 0439 628B          GETSD    EQU    *
439 0440 628B 17 0499        LBSR    CRLF
440 0441 628E 17 045B        LBSR    SETCKL
441 0442 6291 C6 06          LDB     #6
442 0443 6293 8E 6942        LDX     #CMD9
443 0444 6296 A6 80 SC91     LDA     ,X+
444 0445 6298 17 03DC        LBSR    SNDDAT
445 0446 629B 5A              DECB
446 0447 629C 26 F8          BNE     SC91
447 0448          * LBSR CRLF
448 0449          *get response
449 0450 629E          GR90
450 0451 629E 17 03F2        LBSR    GETDAT
451 0452 62A1 B6 6007        LDA     RESDAT
452 0453          * LBSR DSPBIN disp response
453 0454 62A4 81 00          CMPA    #$00
454 0455 62A6 26 F6          BNE     GR90
455 0456
456 0457          * LBSR CRLF
457 0458 62A8 17 047C        LBSR    CRLF
458 0459 62AB C6 14          LDB     #16+1+2+1    fe,crcH,crcL,ff
459 0460 62AD 17 03E3 GR91   LBSR    GETDAT
460 0461 62B0 B6 6007        LDA     RESDAT

```

461	0462	62B3	17	0448	LBSR	DSPBIN	disp response
462	0463	62B6	5A		DECB		
463	0464	62B7	26	F4	BNE	GR91	
464	0465	62B9	17	046B	LBSR	CRLF	
465	0466						
466	0467			*disp offset of MBR,FAT#1,DIR,CLUSTER#2			
467	0468	62BC	8E	6A56	LDX	#MSOFS	
468	0469	62BF	17	051A	LBSR	PSTRNG	
469	0470	62C2	8E	6A66	LDX	#MSBPB	
470	0471	62C5	17	0514	LBSR	PSTRNG	
471	0472	62C8	8E	6009	LDX	#SCTBPB	
472	0473	62CB	C6	04	LDB	#4	
473	0474	62CD	A6	80	SCL0P1	LDA	,X+
474	0475	62CF	17	042C	LBSR	DSPBIN	
475	0476	62D2	5A		DECB		
476	0477	62D3	26	F8	BNE	SCL0P1	
477	0478						
478	0479	62D5	8E	6A6F	LDX	#MSFAT	
479	0480	62D8	17	0501	LBSR	PSTRNG	
480	0481	62DB	8E	600D	LDX	#SCTFAT	
481	0482	62DE	C6	04	LDB	#4	
482	0483	62E0	A6	80	SCL0P2	LDA	,X+
483	0484	62E2	17	0419	LBSR	DSPBIN	
484	0485	62E5	5A		DECB		
485	0486	62E6	26	F8	BNE	SCL0P2	
486	0487						
487	0488	62E8	8E	6A78	LDX	#MSDIR	
488	0489	62EB	17	04EE	LBSR	PSTRNG	
489	0490	62EE	8E	6011	LDX	#SCTDIR	
490	0491	62F1	C6	04	LDB	#4	
491	0492	62F3	A6	80	SCL0P3	LDA	,X+
492	0493	62F5	17	0406	LBSR	DSPBIN	
493	0494	62F8	5A		DECB		
494	0495	62F9	26	F8	BNE	SCL0P3	
495	0496						
496	0497	62FB	8E	6A81	LDX	#MSCL2	
497	0498	62FE	17	04DB	LBSR	PSTRNG	
498	0499	6301	8E	6019	LDX	#SCTCLT	
499	0500	6304	C6	04	LDB	#4	

```

500 0501 6306 A6 80 SCL0P4 LDA ,X+
501 0502 6308 17 03F3 LBSR DSPBIN
502 0503 630B 5A DECB
503 0504 630C 26 F8 BNE SCL0P4
504 0505 630E 17 0416 LBSR CRLF
505 0506 6311 39 RTS
506 0507
507 0508 *
508 0509 6312 SDERR EQU *
509 0510 6312 8E 6A30 LDX #MERR
510 0511 6315 17 04C4 LBSR PSTRNG
511 0512 6318 OKEND
512 0513 6318 39 RTS return to BASIC
513 0514 *
514 0515 *****
515 0516 * READ ROUTINE
516 0517 * use CMD17
517 0518 * set sctno to $6005,6006
518 0519 * read data set to $6800-69FF
519 0520 *****
520 0521 6319 SDRD EQU *
521 0522 6319 17 040B LBSR CRLF
522 0523 631C 17 038E LBSR IN SCT
523 0524 631F 17 02FC LBSR RD1SCT read 1 sct
524 0525
525 0526 *disp data
526 0527 6322 17 0402 LBSR CRLF
527 0528 6325 17 0331 LBSR DSPBUF
528 0529 6328 17 03FC LBSR CRLF
529 0530 632B 39 RTS
530 0531
531 0532 *****
532 0533 * Read MBR & Disp parameter of BPB
533 0534 *****
534 0535 632C DSPMBR EQU *
535 0536 632C 17 03F8 LBSR CRLF
536 0537 632F 8D 19 BSR GETMBR
537 0538
538 0539 *disp BPB

```

```

539 0540 6331 17 03F3          LBSR    CRLF
540 0541 6334 8E 6EBE          LDX     #RWBUF+446
541 0542 6337 C6 10 GET1      LDB     #16
542 0543 6339 A6 80 GET2      LDA     ,X+
543 0544 633B 17 03C0          LBSR    DSPBIN      disp data
544 0545 633E 5A              DECB
545 0546 633F 26 F8          BNE     GET2
546 0547 6341 17 03E3          LBSR    CRLF
547 0548 6344 8C 6EFE          CMPX    #RWBUF+510      446+16*4
548 0549 6347 25 EE          BLO     GET1
549 0550 6349 39              RTS
550 0551
551 0552          *-----
552 0553          *read MBR
553 0554 634A          GETMBR EQU      *
554 0555 634A 8E 6003          LDX     #SCTN0
555 0556 634D C6 04          LDB     #4
556 0557 634F 4F              CLRA
557 0558 6350 A7 80 GPL0P      STA     ,X+
558 0559 6352 5A              DECB
559 0560 6353 26 FB          BNE     GPL0P
560 0561 6355 17 035D          LBSR    SETSCT
561 0562
562 0563 6358 17 02C3          LBSR    RD1SCT      read 1 sct
563 0564 635B 39              RTS
564 0565
565 0566          *-----
566 0567          *set sector no to SCTN0
567 0568          *from X,X+1,X+2,X+3 to SCTN0-SCTN3
568 0569 635C          SETSNO EQU      *
569 0570 635C 108E 6003          LDY     #SCTN0
570 0571 6360 C6 04          LDB     #4
571 0572 6362 A6 80 SEL0P      LDA     ,X+
572 0573 6364 A7 A0          STA     ,Y+
573 0574 6366 5A              DECB
574 0575 6367 26 F9          BNE     SEL0P
575 0576 6369 17 0349          LBSR    SETSCT
576 0577 636C 39              RTS
577 0578

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```

578 0579 *-----
579 0580 *read BPB
580 0581 636D GETBPB EQU *
581 0582 636D 8D DB BSR GETMBR
582 0583 *set sct no of BPB
583 0584 636F 8E 6ECA LDX #RWBUF+446+8+4
584 0585 6372 108E 6003 LDY #SCTN0
585 0586 6376 C6 04 LDB #4
586 0587 6378 A6 82 GPL0P2 LDA , -X
587 0588 637A A7 A0 STA ,Y+
588 0589 637C 5A DECB
589 0590 637D 26 F9 BNE GPL0P2
590 0591 637F 17 0333 LBSR SETSCT
591 0592
592 0593 6382 17 0299 LBSR RD1SCT read 1 sct
593 0594 6385 39 RTS
594 0595
595 0596 *****
596 0597 * Disp BPB(Bios Parameter Block)
597 0598 *****
598 0599 6386 DSPBPB EQU *
599 0600 6386 8E 6009 LDX #SCTBPB
600 0601 6389 8D D1 BSR SETSNO
601 0602 638B 17 0290 LBSR RD1SCT read 1 sct
602 0603
603 0604 *disp BPB
604 0605 638E 17 0396 LBSR CRLF
605 0606 6391 17 02C5 LBSR DSPBUF
606 0607 6394 17 0390 LBSR CRLF
607 0608 6397 39 RTS
608 0609
609 0610 *****
610 0611 * Disp FAT#1
611 0612 *****
612 0613 6398 DSPFAT EQU *
613 0614 6398 8E 600D LDX #SCTFAT
614 0615 639B 8D BF BSR SETSNO
615 0616 639D 17 027E LBSR RD1SCT read 1 sct
616 0617

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```

617      0618      *disp FAT
618      0619  63A0  17  0384          LBSR    CRLF
619      0620  63A3  17  02B3          LBSR    DSPBUF
620      0621  63A6  17  037E          LBSR    CRLF
621      0622  63A9  39                  RTS
622      0623
623      0624      *****
624      0625      * Disp Directory
625      0626      *****
626      0627  63AA      DSPDIR  EQU      *
627      0628  63AA  7C  604E          INC      DPGFLG      set 1st page of DIR
628      flag
629      0629      *return no of DIR
630      0630  63AD  108E  6019          LDY      #SCTDSV+4
631      0631  63B1  8E  6015          LDX      #SCTDIR+4
632      0632  63B4  C6  04          LDB      #4
633      0633  63B6  A6  A2  DRET0  LDA      , -Y
634      0634  63B8  A7  82          STA      , -X
635      0635  63BA  5A          DECB
636      0636  63BB  26  F9          BNE      DRET0
637      0637
638      0638  63BD  8E  6011  DSPFL0  LDX      #SCTDIR
639      0639  63C0  8D  9A          BSR      SETSNO
640      0640  63C2  17  0259          LBSR      RD1SCT      read 1 sct
641      0641
642      0642      *disp FILE name
643      0643  63C5  17  035F          LBSR      CRLF
644      0644  63C8  17  035C          LBSR      CRLF
645      0645  63CB  8E  6D00          LDX      #RWBUF
646      0646  63CE  7D  604E          TST      DPGFLG      1st page of DIR?
647      0647  63D1  27  03          BEQ      DSKP1
648      0648  63D3  30  88  60          LEAX     96,X      skip top data
649      0649  63D6      DSKP1
650      0650
651      0651  63D6  A6  84  DSPFL1  LDA      ,X
652      0652  63D8  1027  008F          LBEQ     DSPEND      if top of
653      filename is 0 then end
654      0653  63DC  81  E5          CMPA     #$E5      delete mark?
655      0654  63DE  27  0A          BEQ      DSPCKE

```

656	0655	63E0	A6	0B		LDA	11,X	attr
657	0656	63E2	81	0F		CMPA	#\$0F	long file name?
658	0657	63E4	27	04		BEQ	DSPCKE	
659	0658	63E6	81	10		CMPA	#\$10	directory?
660	0659	63E8	26	0A		BNE	DSPFL2	
661	0660	63EA	30	88 20	DSPCKE	LEAX	32,X	next directory
662	entry							
663	0661	63ED	8C	6EE0		CMPX	#RWBUF+512-32	
664	0662	63F0	24	60		BHS	DSPFL5	
665	0663	63F2	20	E2		BRA	DSPFL1	
666	0664							
667	0665	63F4			DSPFL2			
668	0666				*disp filename			
669	0667	63F4	C6	08		LDB	#8	
670	0668	63F6	A6	80	DSPFL3	LDA	,X+	
671	0669	63F8	BD	FB BB		JSR	OUTCHA	
672	0670	63FB	5A			DECB		
673	0671	63FC	26	F8		BNE	DSPFL3	
674	0672				*disp ext			
675	0673	63FE	86	2E		LDA	#'.	
676	0674	6400	BD	FB BB		JSR	OUTCHA	
677	0675	6403	C6	03		LDB	#3	
678	0676	6405	A6	80	DSPFL4	LDA	,X+	
679	0677	6407	BD	FB BB		JSR	OUTCHA	
680	0678	640A	5A			DECB		
681	0679	640B	26	F8		BNE	DSPFL4	
682	0680	640D	86	20		LDA	#'	
683	0681	640F	BD	FB BB		JSR	OUTCHA	
684	0682	6412	BD	FB BB		JSR	OUTCHA	
685	0683							
686	0684				*get FILE time data			
687	0685	6415	30	07		LEAX	7,X	
688	0686	6417	EC	81		LDD	,X++	
689	0687	6419	FD	6058		STD	WK1	
690	0688							
691	0689				*get CLUSTER(H) data			
692	0690	641C	EC	81		LDD	,X++	
693	0691	641E	1E	89		EXG	A,B	
694	0692	6420	FD	6047		STD	CLSTH	

695	0693	6423	30	04		LEAX	4,X	
696	0694	6425	EC	81		LDD	,X++	
697	0695	6427	1E	89		EXG	A,B	
698	0696	6429	FD	6049		STD	CLSTL	
699	0697							
700	0698					*get FILE size data		
701	0699	642C	108E	6045		LDY	#WFSIZ+4	
702	0700	6430	C6	04		LDB	#4	
703	0701	6432	A6	80	DSPC1	LDA	,X+	
704	0702	6434	A7	A2		STA	, -Y	
705	0703	6436	5A			DECB		
706	0704	6437	26	F9		BNE	DSPC1	
707	0705							
708	0706					*disp FILE size		
709	0707	6439	86	20		LDA	#'	
710	0708	643B	BD	FB BB		JSR	OUTCHA	
711	0709	643E	17	040C		LBSR	DSPSIZ	disp FILE size
712	0710							
713	0711					*disp FILE time		
714	0712	6441	FC	6058		LDD	WK1	
715	0713	6444	17	03A5		LBSR	DSPYMD	disp MM-DD-YY
716	0714							
717	0715					*disp CLUSTER(H,L)		
718	0716	6447	17	0462		LBSR	DSPCLS	
719	0717	644A	17	02DA		LBSR	CRLF	
720	0718							
721	0719	644D	8C	6F00		CMPX	#RWBUF+512	
722	0720	6450	25	84		BLO	DSPFL1	
723	0721	6452	8E	6011	DSPFL5	LDX	#SCTDIR	
724	0722	6455	17	0192		LBSR	INCX4	sct no +1
725	0723	6458	7F	604E		CLR	DPGFLG	clear 1st page of
726	DIR							
727	0724	645B	8E	6AC1		LDX	#MSNXTTP	
728	0725	645E	17	037B		LBSR	PSTRNG	
729	0726	6461	BD	690B		JSR	INOUTC	
730	0727	6464	81	20		CMPA	#'	
731	0728	6466	26	03		BNE	DSPEND	
732	0729	6468	16	FF52		LBRA	DSPFL0	
733	0730	646B	17	02B9	DSPEND	LBSR	CRLF	

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734 0731 646E 39          RTS
735 0732
736 0733 *****
737 0734 * Load FILE
738 0735 *****
739 0736 646F      LDFILE EQU      *
740 0737      * LBSR DSPDIR
741 0738 646F 7C 604E          INC      DPGFLG      set 1st page of DIR
742 flag
743 0739      *return no of DIR
744 0740 6472 108E 6019          LDY      #SCTDSV+4
745 0741 6476 8E 6015          LDX      #SCTDIR+4
746 0742 6479 C6 04          LDB      #4
747 0743 647B A6 A2 DRET1 LDA      , -Y
748 0744 647D A7 82          STA      , -X
749 0745 647F 5A          DECB
750 0746 6480 26 F9          BNE      DRET1
751 0747
752 0748 6482 8E 6A91          LDX      #MSINFN
753 0749 6485 17 0354          LBSR      PSTRNG
754 0750 6488 17 0467          LBSR      FNMIN
755 0751
756 0752 648B 8E 6011 LDFL0 LDX      #SCTDIR
757 0753 648E 17 FECB          LBSR      SETSNO
758 0754 6491 17 018A          LBSR      RD1SCT      read 1 sct
759 0755
760 0756 6494 8E 6D00          LDX      #RWBUF
761 0757 6497 7D 604E          TST      DPGFLG      1st page of DIR?
762 0758 649A 27 03          BEQ      LDFL1
763 0759 649C 30 88 60          LEAX      96,X      skip top data
764 0760
765 0761 649F A6 84 LDFL1 LDA      ,X
766 0762 64A1 1027 00FA          LBEQ      LDNEND      if top of
767 filename is 0 then end
768 0763 64A5 81 E5          CMPA      #$E5      delete mark?
769 0764 64A7 26 0A          BNE      LD1
770 0765 64A9 30 88 20          LEAX      32,X      next directory
771 entry
772 0766 64AC 8C 6F00          CMPX      #RWBUF+512      over buf?

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773	0767	64AF	24	0F		BHS	LDFL5	
774	0768	64B1	20	EC		BRA	LDFL1	
775	0769							
776	0770				*check file name			
777	0771	64B3			LD1			
778	0772	64B3	17	041B		LBSR	CMPFN	C=1:match
779	0773	64B6	25	13		BCS	LD2	
780	0774	64B8	30	88 20		LEAX	32,X	
781	0775	64BB	8C	6F00		CMPX	#RWBUF+512	
782	0776	64BE	25	DF		BLO	LDFL1	
783	0777							
784	0778	64C0			LDFL5			
785	0779	64C0	8E	6011		LDX	#SCTDIR	
786	0780	64C3	17	0124		LBSR	INCX4	sct no +1
787	0781	64C6	7F	604E		CLR	DPGFLG	clear 1st page of
788	DIR							
789	0782	64C9	20	C0		BRA	LDFL0	
790	0783							
791	0784				*hit file, get dirctory entry			
792	0785	64CB	C6	20	LD2	LDB	#32	
793	0786	64CD	108E	601E		LDY	#DIRENT	
794	0787	64D1	A6	80	DCPY	LDA	,X+	
795	0788	64D3	A7	A0		STA	,Y+	
796	0789	64D5	5A			DECB		
797	0790	64D6	26	F9		BNE	DCPY	
798	0791	64D8	8E	601E		LDX	#DIRENT	
799	0792							
800	0793				*get cluster no			
801	0794	64DB	EC	88 14		LDD	\$14,X	
802	0795	64DE	1E	89		EXG	A,B	
803	0796	64E0	FD	6047		STD	CLSTH	high cluster no
804	0797	64E3	EC	88 1A		LDD	\$1A,X	
805	0798	64E6	1E	89		EXG	A,B	
806	0799	64E8	FD	6049		STD	CLSTL	low cluster no
807	0800				*sub start cluster no (SD:2,SDHC:3)			
808	0801	64EB	83	0002		SUBD	#2	
809	0802	64EE	7D	6008		TST	SDHC	SD:0,SDHC:1
810	0803	64F1	27	03		BEQ	LD3	
811								

812	0804	64F3	83	0001		SUBD	#1	D:cluster no (start
813		0)						
814		0805						
815	0806	64F6	34	36	LD3	PSHS	A,B,X,Y	
816	0807	64F8	8E	6019		LDX	#SCTCLT	
817	0808	64FB	108E	6058		LDY	#WK1	
818	0809	64FF	C6	04		LDB	#4	
819	0810	6501	A6	80	LD4	LDA	,X+	
820	0811	6503	A7	A0		STA	,Y+	
821	0812	6505	5A			DECB		
822	0813	6506	26	F9		BNE	LD4	
823	0814	6508	35	36		PULS	A,B,X,Y	
824		0815						
825	0816						*(B)x40h	
826	0817	650A	34	30		PSHS	X,Y	
827	0818	650C	8E	6058		LDX	#WK1	
828	0819	650F	17	00E7		LBSR	ADD40H	
829	0820	6512	35	30		PULS	X,Y	
830		0821						
831	0822						*get file size	
832	0823	6514	34	30		PSHS	X,Y	
833	0824	6516	30	88 20		LEAX	32,X	
834	0825	6519	108E	6041		LDY	#WFSIZ	
835	0826	651D	C6	04		LDB	#4	
836	0827	651F	A6	82	LD41	LDA	, -X	
837	0828	6521	A7	A0		STA	,Y+	
838	0829	6523	5A			DECB		
839	0830	6524	26	F9		BNE	LD41	
840	0831	6526	35	30		PULS	X,Y	
841		0832						
842	0833	6528	EC	88 1D		LDD	\$1D,X	upper of file size
843	0834	652B	1E	89		EXG	A,B	
844	0835	652D	44			LSRA	wfsiz/\$200+1	
845	0836	652E	56			RORB		
846	0837	652F	C3	0001		ADDD	#1	
847		0838						
848	0839	6532	C1	40		CMPB	#\$40	40h page?
849	0840	6534	23	02		BLS	LD5	
850	0841	6536	C6	40		LDB	#\$40	

851	0842	6538	F7	604B	LD5	STB	CNT	sct counter
852	0843	653B	F7	604C		STB	CNTSV	
853	0844							
854	0845							*set start sct no
855	0846	653E			LD6			
856	0847	653E	34	30		PSHS	X,Y	
857	0848	6540	8E	6058		LDX	#WK1	
858	0849	6543	108E	6003		LDY	#SCTN0	
859	0850	6547	C6	04		LDB	#4	
860	0851	6549	A6	80	LD7	LDA	,X+	
861	0852	654B	A7	A0		STA	,Y+	
862	0853	654D	5A			DECB		
863	0854	654E	26	F9		BNE	LD7	
864	0855	6550	35	30		PULS	X,Y	
865	0856							
866	0857	6552	17	0160		LBSR	SETSCT	
867	0858	6555	17	00C6		LBSR	RD1SCT	read 1 file data
868	0859							
869	0860	6558	7F	604D		CLR	MCFLG	set not machine
870	code							
871	0861	655B	B6	604C		LDA	CNTSV	
872	0862	655E	B1	604B		CMPA	CNT	top sector read?
873	0863	6561	26	1B		BNE	LD8	
874	0864							
875	0865							*at top sector, ini set
876	0866							*offset 00 data is
877	0867							* \$00 :machine code (MCFLG=0)
878	0868							* \$FF,\$FE:BASIC PROGRAM (binary save) (MCFLG=\$FF,\$FE)
879	0869							* else :general text (MCFLG=1)
880	0870	6563	10BE	6043		LDY	WFSIZ+2	
881	0871	6567	10BF	6045		STY	FILSIZ	set file size
882	0872							
883	0873	656B	8E	6D00		LDX	#RWBUF	
884	0874	656E	A6	84		LDA	,X	00?(machine code?)
885	0875	6570	B7	604D		STA	MCFLG	
886	0876	6573	26	09		BNE	LD8	
887	0877							*at machine code, read header
888	0878							* INC MCFLG set machine code
889	0879	6575	10AE	01		LDY	1,X	get file size

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890 0880 6578 10BF 6045          STY      FILSIZ
891 0881 657C EE 03          LDU      3,X          get store addr
892 0882
893 0883 657E DE 33 LD8      LDU      BASADRP      get text buffer adr
894 0884 6580 4D          TSTA
895 0885 6581 2B 03          BMI      LD9          $FFor$FE (basic
896 code?)
897 0886 6583 CE 2000          LDU      #TXTADR      (else)
898 0887 6586          LD9
899 0888 6586 8D 1F          BSR      MEMSET
900 0889
901 0890 6588 8E 6058          LDX      #WK1
902 0891 658B 8D 5D          BSR      INCX4      sct no +1
903 0892 658D 7F 604E          CLR      DPGFLG      clear 1st page of
904 DIR
905 0893 6590 7A 604B          DEC      CNT
906 0894 6593 26 A9          BNE      LD6
907 0895
908 0896 6595 8E 6AA7 LDEND  LDX      #MSLDED
909 0897 6598 17 0241          LBSR      PSTRNG
910 0898 659B 17 0189 LDE1  LBSR      CRLF
911 0899 659E 39          RTS
912 0900 659F 8E 6AB3 LDNEND  LDX      #MSNOFL
913 0901 65A2 17 0237          LBSR      PSTRNG
914 0902 65A5 20 F4          BRA      LDE1
915 0903
916 0904          *
917 0905          *data to memory
918 0906          *U:memory adr
919 0907
920 0908 65A7          MEMSET EQU      *
921 0909 65A7 34 30          PSHS      X,Y
922 0910 65A9 8E 6D00          LDX      #RWBUF
923 0911 65AC B6 604D          LDA      MCFLG      0,1,$FF($FE)
924 0912 65AF 27 07          BEQ      MS1
925 0913 65B1 2B 12          BMI      MS2
926 0914          *text file
927 0915 65B3 CE 2000          LDU      #TXTADR      (else)
928 0916 65B6 20 1B          BRA      MSLOP0

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```

929 0917      *machine code
930 0918 65B8      MS1
931 0919 65B8 10AE 01      LDY      1,X      get file size
932 0920 65BB 10BF 6045      STY      FILSIZ
933 0921 65BF EE 03      LDU      3,X      get store addr
934 0922 65C1 30 05      LEAX      5,X      skip header
935 0923 65C3 20 0E      BRA      MSLOP0
936 0924      *basic program(binary save)
937 0925 65C5      MS2
938 0926 65C5 30 03      LEAX      3,X      skip 3 $FFs
939 0927 65C7 10BE 6045      LDY      FILSIZ
940 0928 65CB 31 3D      LEAY      -3,Y
941 0929 65CD 10BF 6045      STY      FILSIZ
942 0930 65D1 DE 33      LDU      BASADRP      get text buffer adrs
943 0931
944 0932 65D3 10BE 6045 MSLOP0 LDY      FILSIZ
945 0933 65D7 A6 80 MSLOP LDA      ,X+
946 0934 65D9 A7 C0      STA      ,U+
947 0935 65DB 31 3F      LEAY      -1,Y
948 0936 65DD 27 05      BEQ      MS3
949 0937 65DF 8C 6F00      CMPX      #RWBUF+512
950 0938 65E2 25 F3      BLO      MSLOP
951 0939 65E4 10BF 6045 MS3 STY      FILSIZ
952 0940 65E8 35 B0      PULS      X,Y,PC
953 0941
954 0942      *-----
955 0943      *(X) .. (X+3)+1
956 0944 65EA      INCX4 EQU      *
957 0945 65EA 6C 03 ADX0 INC      3,X
958 0946 65EC 24 02      BCC      ADX1
959 0947 65EE 6C 02      INC      2,X
960 0948 65F0 24 02 ADX1 BCC      ADX2
961 0949 65F2 6C 01      INC      1,X
962 0950 65F4 24 02 ADX2 BCC      ADX3
963 0951 65F6 6C 84      INC      ,X
964 0952 65F8 39      ADX3 RTS
965 0953      *
966 0954      *(X) .. (X+3)+(B)x40h
967 0955 65F9      ADD40H EQU      *

```

968	0956	65F9	34	06		PSHS	A,B
969	0957	65FB	A6	03	ADD0	LDA	3,X
970	0958	65FD	8B	40		ADDA	#\$40
971	0959	65FF	A7	03		STA	3,X
972	0960	6601	24	06		BCC	ADD1
973	0961	6603	A6	02		LDA	2,X
974	0962	6605	8B	01		ADDA	#1
975	0963	6607	A7	02		STA	2,X
976	0964	6609	24	06	ADD1	BCC	ADD2
977	0965	660B	A6	01		LDA	1,X
978	0966	660D	8B	01		ADDA	#1
979	0967	660F	A7	01		STA	1,X
980	0968	6611	24	06	ADD2	BCC	ADD3
981	0969	6613	A6	84		LDA	,X
982	0970	6615	8B	01		ADDA	#1
983	0971	6617	A7	84		STA	,X
984	0972	6619	5A		ADD3	DECB	
985	0973	661A	26	DF		BNE	ADD0
986	0974	661C	35	86		PULS	A,B,PC
987	0975						
988	0976		*				
989	0977		*-----				
990	0978		*read 1 sct				
991	0979		*-----				
992	0980	661E			RD1SCT EQU	*	
993	0981				*copy sct no to cmd param		
994	0982	661E	8E	6003		LDX	#SCTN0
995	0983	6621	108E	691F		LDY	#CMD17+1
996	0984	6625	C6	04		LDB	#4
997	0985	6627	A6	80	CPY1	LDA	,X+
998	0986	6629	A7	A0		STA	,Y+
999	0987	662B	5A			DECB	
1000	0988	662C	26	F9		BNE	CPY1
1001	0989				*send cmd		
1002	0990	662E	17	00BB		LBSR	SETCKL
1003	0991	6631	C6	06		LDB	#6
1004	0992	6633	8E	691E		LDX	#CMD17
1005	0993	6636	A6	80	RD1	LDA	,X+
1006	0994	6638	8D	3D		BSR	SNDDAT

```

1007 0995 663A 5A          DECB
1008 0996 663B 26 F9      BNE      RD1
1009 0997          *get response
1010 0998 663D          RD2
1011 0999 663D 8D 54      BSR      GETDAT
1012 1000 663F B6 6007      LDA      RESDAT
1013 1001 6642 81 FE      CMPA     #$FE      token?
1014 1002 6644 26 F7      BNE      RD2
1015 1003
1016 1004          *get data
1017 1005          * LBSR CRLF
1018 1006 6646 8E 6D00      LDX      #RWBUF      <-temp -1=""
1019 0094="" 00b3="" 00cb="" 0200="" 02="" 06="" 08="" 1007="" 1008=""
1020 1009="" 1010="" 1011="" 1012="" 1013="" 1014="" 1015="" 1016=""
1021 1017="" 1018="" 1019="" 1020="" 1021="" 1022="" 1023="" 1024=""
1022 1025="" 1026="" 1027="" 1028="" 1029="" 1030="" 1031="" 1032=""
1023 1033="" 1034="" 1035="" 1036="" 1037="" 1038="" 1039="" 1040=""
1024 1041="" 1042="" 1043="" 1044="" 1045="" 1046="" 1047="" 1048=""
1025 1049="" 1050="" 1051="" 1052="" 1053="" 1054="" 1055="" 1056=""
1026 1057="" 1058="" 1059="" 1060="" 1061="" 1062="" 1063="" 108e="" 12=""
1027 17="" 20="" 26="" 27="" 31="" 34="" 35="" 39="" 3f="" 44="" 48=""
1028 53="" 57="" 5a="" 6007="" 63="" 6649="" 664d="" 664f="" 6652=""
1029 6654="" 6656="" 6658="" 6659="" 665c="" 665f="" 6663="" 6665=""
1030 6667="" 666a="" 666c="" 666e="" 666f="" 6671="" 6674="" 6676=""
1031 6677="" 6679="" 667c="" 667e="" 6680="" 6682="" 6684="" 6687=""
1032 6689="" 668b="" 668c="" 668e="" 668f="" 6691="" 6693="" 6695=""
1033 6697="" 6699="" 669a="" 669b="" 669c="" 669e="" 66a1="" 66a2="" 67=""
1034 6d00="" 7f="" 80="" 84="" 86="" 8a="" 8d="" 8e="" a6="" a7="" a=""
1035 anda="" b1="" b6="" b7="" beq="" bne="" bra="" bsr="" buf="" c6=""
1036 clr="" crlf="" cs:b0="L" data="" decb="" disp="" dspb1="" dspb2=""
1037 dspb3="" dspbin="" dspbuf="" ed="" equ="" f4="" f5="" fc="" fd9e=""
1038 get="" getd1="" getdat="" lbsr="" lda="" ldb="" ldx="" ldy="" leay=""
1039 ls1a="" lsra="" miso:b1="H(for" nop="" ora="" pbdatt="" pia="" pshs=""
1040 puls="" rd3="" resdat="" response="" rts="" save="" send="" setckh=""
1041 setckl="" sndd1="" snddat="" sta="" temp="">C
1042 1064 66A3 79 6007      ROL      RESDAT      C->b0
1043 1065 66A6 5A          DECB
1044 1066 66A7 26 EE      BNE      GETD1
1045 1067 66A9 8D 41      BSR      SETCKL

```

1046	1068	66AB	35	86		PULS	A,B,PC	
1047	1069		*					
1048	1070		*-----					
1049	1071		*input SCT No					
1050	1072	66AD		INSCT	EQU	*		
1051	1073	66AD	8E	69E6		LDX	#MSSCT	
1052	1074	66B0	17	0129		LBSR	PSTRNG	
1053	1075	66B3	8D	7D		BSR	GET8CH	get 8chr, convert
1054	4bin							
1055	1076							
1056	1077	66B5		SETSCT	EQU	*		
1057	1078	66B5	34	32		PSHS	A,X,Y	
1058	1079	66B7	7D	6008		TST	SDHC	
1059	1080	66BA	26	1E		BNE	CALCHC	
1060	1081		*SD card					
1061	1082	66BC	C6	03		LDB	#3	
1062	1083	66BE	8E	6004		LDX	#SCTN1	
1063	1084	66C1	108E	6003		LDY	#SCTN0	
1064	1085	66C5	A6	80	SFT1	LDA	,X+	
1065	1086	66C7	A7	A0		STA	,Y+	
1066	1087	66C9	5A			DECB		
1067	1088	66CA	26	F9		BNE	SFT1	
1068	1089	66CC	7F	6006		CLR	SCTN3	
1069	1090		*calc top adrs of sct					
1070	1091	66CF		CALCSD	EQU	*		
1071	1092	66CF	78	6006		LSL	SCTN3	
1072	1093	66D2	79	6005		ROL	SCTN2	
1073	1094	66D5	79	6004		ROL	SCTN1	
1074	1095	66D8	35	B2		PULS	A,X,Y,PC	
1075	1096							
1076	1097		*SDHC card					
1077	1098		*calc top adrs of sct					
1078	1099	66DA		CALCHC	EQU	*		
1079	1100	66DA	8E	6003		LDX	#SCTN0	
1080	1101	66DD	108E	691F		LDY	#CMD17+1	
1081	1102	66E1	C6	04		LDB	#4	
1082	1103	66E3	A6	80	CALCH1	LDA	,X+	
1083	1104	66E5	A7	A0		STA	,Y+	
1084	1105	66E7	5A			DECB		

1085	1106	66E8	26	F9	BNE	CALCH1	
1086	1107	66EA	35	B2	PULS	A,X,Y,PC	
1087	1108						
1088	1109		*				
1089	1110		*set sck:L				
1090	1111	66EC		SETCKL	EQU	*	
1091	1112	66EC	34	02	PSHS	A	
1092	1113	66EE	86	34	LDA	#\$34	ca2=L
1093	1114	66F0	B7	FD9F	STA	PBCTL	
1094	1115	66F3	35	82	PULS	A,PC	
1095	1116						
1096	1117		*set sck:H				
1097	1118	66F5		SETCKH	EQU	*	
1098	1119	66F5	34	02	PSHS	A	
1099	1120	66F7	86	3C	LDA	#\$3C	ca2=H
1100	1121	66F9	B7	FD9F	STA	PBCTL	
1101	1122	66FC	35	82	PULS	A,PC	
1102	1123						
1103	1124		*binary(A)->ascii(A)(B)				
1104	1125	66FE		DSPBIN	EQU	*	
1105	1126	66FE	34	06	PSHS	A,B	
1106	1127	6700	1F	89	TFR	A,B	
1107	1128	6702	44		LSRA		
1108	1129	6703	44		LSRA		
1109	1130	6704	44		LSRA		
1110	1131	6705	44		LSRA		
1111	1132	6706	8A	30	ORA	#\$30	
1112	1133	6708	C4	0F	ANDB	#\$0F	
1113	1134	670A	CA	30	ORB	#\$30	
1114	1135	670C	81	39	CMPA	#\$39	
1115	1136	670E	23	02	BLS	BI21	
1116	1137	6710	8B	07	ADDA	#\$07	
1117	1138	6712	C1	39	BI21	CMPB	#\$39
1118	1139	6714	23	02	BLS	BI22	
1119	1140	6716	CB	07	ADDB	#\$07	
1120	1141	6718	BD	FB BB	BI22	JSR	OUTCHA
1121	1142	671B	1F	98	TFR	B,A	
1122	1143	671D	BD	FB BB	JSR	OUTCHA	
1123	1144	6720	86	20	LDA	#\$20	

1124	1145	6722	BD	FBBB		JSR	OUTCHA
1125	1146	6725	35	86		PULS	A,B,PC
1126	1147						
1127	1148		*crlf				
1128	1149	6727		CRLF	EQU	*	
1129	1150	6727	86	0D		LDA	#\$0D
1130	1151	6729	BD	FBBB		JSR	OUTCHA
1131	1152	672C	86	0A		LDA	#\$0A
1132	1153	672E	BD	FBBB		JSR	OUTCHA
1133	1154	6731	39			RTS	
1134	1155						
1135	1156		*				
1136	1157	6732		GET8CH	EQU	*	
1137	1158		*input chr				
1138	1159	6732	8E	6ADA		LDX	#INPBUF
1139	1160	6735	5F			CLRB	
1140	1161	6736	BD	690B	CLOP	JSR	INOUTC
1141	1162	6739	81	0D		CMPA	#\$0D
1142	1163	673B	27	07		BEQ	INEND
1143	1164	673D	A7	85		STA	B,X
1144	1165	673F	5C			INCB	
1145	1166	6740	C1	08		CMPB	#8
1146	1167	6742	25	F2		BLO	CLOP
1147	1168						
1148	1169	6744		INEND			
1149	1170		*shift				
1150	1171	6744	5D			TSTB	
1151	1172	6745	27	11		BEQ	CL1
1152	1173	6747	8E	6ADA		LDX	#INPBUF
1153	1174	674A	3A			ABX	
1154	1175	674B	108E	6AE2		LDY	#INPBUF+8
1155	1176	674F	34	04		PSHS	B
1156	1177	6751	A6	82	SFLOP	LDA	, -X
1157	1178	6753	A7	A2		STA	, -Y
1158	1179	6755	5A			DECB	
1159	1180	6756	26	F9		BNE	SFLOP
1160	1181		*clr				
1161	1182	6758	C6	08	CL1	LDB	#8
1162	1183	675A	E0	E0		SUBB	,S+

1163	1184	675C	27	0A	BEQ	CLEND
1164	1185	675E	8E	6ADA	LDX	#INPBUF
1165	1186	6761	86	30	LDA	#'0
1166	1187	6763	A7	80	CLLOP STA	,X+
1167	1188	6765	5A		DECB	
1168	1189	6766	26	FB	BNE	CLLOP
1169	1190	6768			CLEND	
1170	1191					
1171	1192				*convert to bin	
1172	1193	6768	C6	04	LDB	#4
1173	1194	676A	8E	6ADA	LDX	#INPBUF
1174	1195	676D	108E	6003	LDY	#SCTN0
1175	1196	6771			CVT1	
1176	1197	6771	A6	80	LDA	,X+
1177	1198	6773	8D	1E	BSR	CVBIN
1178	1199	6775	25	16	BCS	EREND1
1179	1200	6777	48		LSLA	
1180	1201	6778	48		LSLA	
1181	1202	6779	48		LSLA	
1182	1203	677A	48		LSLA	
1183	1204	677B	34	02	PSHS	A
1184	1205	677D	A6	80	LDA	,X+
1185	1206	677F	8D	12	BSR	CVBIN
1186	1207	6781	25	08	BCS	EREND0
1187	1208	6783	AB	E0	ADDA	,S+
1188	1209	6785	A7	A0	STA	,Y+
1189	1210	6787	5A		DECB	
1190	1211	6788	26	E7	BNE	CVT1
1191	1212	678A	39		RTS	
1192	1213					
1193	1214	678B	35	02	EREND0 PULS	A
1194	1215	678D	8E	6A47	EREND1 LDX	#MSERR
1195	1216	6790	8D	4A	BSR	PSTRNG
1196	1217	6792	39	EEND	RTS	
1197	1218					
1198	1219				*-----	
1199	1220				*ascii -> bin	
1200	1221	6793		CVBIN	EQU	*
1201	1222	6793	80	30	SUBA	#'0

1202	1223	6795	2B	11		BMI	CVERR	
1203	1224	6797	81	09		CMPA	#9	
1204	1225	6799	23	0A		BLS	CVBEND	
1205	1226	679B	81	11		CMPA	#\$11	
1206	1227	679D	25	09		BLO	CVERR	
1207	1228	679F	81	16		CMPA	#\$16	
1208	1229	67A1	22	05		BHI	CVERR	
1209	1230	67A3	80	07		SUBA	#7	
1210	1231	67A5			CVBEND			
1211	1232	67A5	1C	FE		ANDCC	#\$FE	reset C
1212	1233	67A7	39			RTS		
1213	1234	67A8	1A	01	CVERR	ORCC	#\$01	set C
1214	1235	67AA	39			RTS		
1215	1236							
1216	1237				*-----			
1217	1238				*add 2bytes to 4bytes			
1218	1239				* X,X+1,X+2,X+3 + Y,Y+1			
1219	1240				* data X is big endian			
1220	1241				* data Y is little endian			
1221	1242	67AB			ADD42 EQU *			
1222	1243	67AB	34	36		PSHS	A,B,X,Y	
1223	1244	67AD	32	7C		LEAS	-4,S	
1224	1245	67AF	C6	04		LDB	#4	
1225	1246	67B1	A6	80	AD1	LDA	,X+	
1226	1247	67B3	A7	E0		STA	,S+	
1227	1248	67B5	5A			DECB		
1228	1249	67B6	26	F9		BNE	AD1	
1229	1250							
1230	1251				*add (lower)			
1231	1252	67B8	A6	A0		LDA	,Y+	
1232	1253	67BA	AB	E2		ADDA	, -S	
1233	1254	67BC	A7	E4		STA	,S	
1234	1255	67BE	24	02		BCC	AD2	
1235	1256	67C0	6C	7F		INC	-1,S	
1236	1257				*add (upper)			
1237	1258	67C2	A6	A4	AD2	LDA	,Y	
1238	1259	67C4	AB	E2		ADDA	, -S	
1239	1260	67C6	A7	E4		STA	,S	
1240	1261	67C8	24	02		BCC	AD3	

1241	1262	67CA	6C	7F		INC	-1,S
1242	1263	67CC	30	1C	AD3	LEAX	-4,X
1243	1264	67CE	32	7E		LEAS	-2,S
1244	1265	67D0	C6	04		LDB	#4
1245	1266	67D2	A6	E0	AD4	LDA	,S+
1246	1267	67D4	A7	80		STA	,X+
1247	1268	67D6	5A			DECB	
1248	1269	67D7	26	F9		BNE	AD4
1249	1270	67D9	35	36		PULS	A,B,X,Y
1250	1271	67DB	39			RTS	
1251	1272						
1252	1273					*-----	
1253	1274					* pstring	
1254	1275	67DC			PSTRNG EQU	*	
1255	1276	67DC	34	02		PSHS	A
1256	1277	67DE	A6	80	PSLOP	LDA	,X+
1257	1278	67E0	81	04		CMPA	#4
1258	1279	67E2	27	05		BEQ	PSEND
1259	1280	67E4	BD	FBBB		JSR	OUTCHA
1260	1281	67E7	20	F5		BRA	PSLOP
1261	1282	67E9	35	02	PSEND	PULS	A
1262	1283	67EB	39			RTS	
1263	1284						
1264	1285					*-----	
1265	1286					*disp MM-DD-YY	
1266	1287					*indata A(Low),B(High)	
1267	1288	67EC			DSPYMD EQU	*	
1268	1289	67EC	1E	89		EXG	A,B
1269	1290	67EE	34	10		PSHS	X
1270	1291	67F0	34	06		PSHS	A,B
1271	1292	67F2	C4	E0		ANDB	#\$E0
1272	1293	67F4	54			LSRB	
1273	1294	67F5	54			LSRB	
1274	1295	67F6	54			LSRB	
1275	1296	67F7	54			LSRB	
1276	1297	67F8	54			LSRB	
1277	1298	67F9	47			ASRA	
1278	1299	67FA	24	02		BCC	DYMD1
1279	1300	67FC	CB	08		ADDB	#8

1280	1301	67FE	F7	603E	DYMD1	STB	WMDY	MM
1281	1302	6801	35	06		PULS	A,B	
1282	1303	6803	34	04		PSHS	B	
1283	1304	6805	C4	1F		ANDB	#\$1F	
1284	1305	6807	F7	603F		STB	WMDY+1	DD
1285	1306	680A	35	04		PULS	B	
1286	1307							
1287	1308	680C	44			LSRA		
1288	1309		*	ASLB				
1289	1310		*	BCC	DYMD2			
1290	1311		*	INCA				
1291	1312	680D		DYMD2				
1292	1313	680D	80	14		SUBA	#20	A+1980-2000
1293	1314	680F	B7	6040		STA	WMDY+2	YY
1294	1315							
1295	1316							
1296	1317	6812	86	20		LDA	#'	
1297	1318	6814	BD	FB BB		JSR	OUTCHA	
1298	1319	6817	8E	603E		LDX	#WMDY	
1299	1320	681A	A6	80		LDA	,X+	
1300	1321	681C	8D	14		BSR	DB2ASC	MM
1301	1322	681E	86	2D		LDA	#' -	
1302	1323	6820	BD	FB BB		JSR	OUTCHA	
1303	1324	6823	A6	80		LDA	,X+	
1304	1325	6825	8D	0B		BSR	DB2ASC	DD
1305	1326	6827	86	2D		LDA	#' -	
1306	1327	6829	BD	FB BB		JSR	OUTCHA	
1307	1328	682C	A6	80		LDA	,X+	
1308	1329	682E	8D	02		BSR	DB2ASC	YY
1309	1330	6830	35	90		PULS	X,PC	
1310	1331		*					
1311	1332		*	-----				
1312	1333		*	(A:BIN) ->ASCII				
1313	1334	6832		DB2ASC	EQU	*		
1314	1335	6832	34	06		PSHS	A,B	
1315	1336	6834	1F	89		TFR	A,B	
1316	1337	6836	86	30		LDA	#\$30	
1317	1338	6838	C1	0A	DBLOP	CMPB	#10	
1318	1339	683A	25	05		BLO	DB2	

1319	1340	683C	C0	0A		SUBB	#10	
1320	1341	683E	4C			INCA		
1321	1342	683F	20	F7		BRA	DBLOP	
1322	1343	6841	BD	FB BB	DB2	JSR	OUTCHA	
1323	1344	6844	1F	98		TFR	B,A	
1324	1345	6846	8B	30		ADDA	#\$30	
1325	1346	6848	BD	FB BB		JSR	OUTCHA	
1326	1347	684B	35	86		PULS	A,B,PC	
1327	1348							
1328	1349					*disp FILE size		
1329	1350	684D			DSPSIZ EQU	*		
1330	1351	684D	34	10		PSHS	X	
1331	1352	684F	BE	6041		LDX	WFSIZ	
1332	1353	6852	27	16		BEQ	DSS1	if 64k over?
1333	1354	6854	86	3E		LDA	#' >	
1334	1355	6856	BD	FB BB		JSR	OUTCHA	
1335	1356	6859	86	36		LDA	#' 6	
1336	1357	685B	BD	FB BB		JSR	OUTCHA	
1337	1358	685E	86	34		LDA	#' 4	
1338	1359	6860	BD	FB BB		JSR	OUTCHA	
1339	1360	6863	86	6B		LDA	#' k	
1340	1361	6865	BD	FB BB		JSR	OUTCHA	
1341	1362	6868	20	3B		BRA	DSEND	
1342	1363	686A	86	20	DSS1	LDA	#'	
1343	1364	686C	BD	FB BB		JSR	OUTCHA	
1344	1365	686F	B6	6043		LDA	WFSIZ+2	under 64k
1345	1366					*disp dec		
1346	1367	6872	81	64		CPMA	#100	
1347	1368	6874	23	12		BLS	DSS13	
1348	1369	6876	1F	89		TFR	A,B	
1349	1370	6878	86	30		LDA	#\$30	
1350	1371	687A	C1	64	DSL0P1	CM PB	#100	
1351	1372	687C	25	05		BLO	DSS12	
1352	1373	687E	C0	64		SUBB	#100	
1353	1374	6880	4C			INCA		
1354	1375	6881	20	F7		BRA	DSL0P1	
1355	1376	6883	BD	FB BB	DSS12	JSR	OUTCHA	
1356	1377	6886	20	05		BRA	DSS14	
1357	1378					*under 100		

1358	1379	6888	86	20	DSS13	LDA	#'
1359	1380	688A	BD	FB	BB	JSR	OUTCHA
1360	1381	688D	86	30	DSS14	LDA	#\$30
1361	1382	688F	C1	0A	DSL0P2	CMPB	#10
1362	1383	6891	25	05		BL0	DSS2
1363	1384	6893	C0	0A		SUBB	#10
1364	1385	6895	4C			INCA	
1365	1386	6896	20	F7		BRA	DSL0P2
1366	1387	6898	81	30	DSS2	CMPA	#\$30
1367	1388	689A	26	01		BNE	DSS3
1368	1389	689C	4C			INCA	
1369	1390	689D	BD	FB	BB	DSS3 JSR	OUTCHA
1370	1391	68A0	86	6B		LDA	#'k
1371	1392	68A2	BD	FB	BB	JSR	OUTCHA
1372	1393	68A5	86	20	DSEND	LDA	#'
1373	1394	68A7	BD	FB	BB	JSR	OUTCHA
1374	1395	68AA	35	90		PULS	X,PC
1375	1396						
1376	1397				*-----		
1377	1398	68AC			DSPCLS EQU	*	
1378	1399	68AC	34	10		PSHS	X
1379	1400	68AE	86	20		LDA	#'
1380	1401	68B0	BD	FB	BB	JSR	OUTCHA
1381	1402	68B3	8E	6047		LDX	#CLSTH
1382	1403	68B6	C6	02		LDB	#2
1383	1404	68B8	A6	80	DCL0P1	LDA	,X+
1384	1405	68BA	17	FF75		LBSR	DB2ASC
1385	1406	68BD	5A			DECB	
1386	1407	68BE	26	F8		BNE	DCL0P1
1387	1408	68C0	86	20		LDA	#'
1388	1409	68C2	BD	FB	BB	JSR	OUTCHA
1389	1410	68C5	C6	02		LDB	#2
1390	1411	68C7	A6	80	DCL0P2	LDA	,X+
1391	1412	68C9	17	FF66		LBSR	DB2ASC
1392	1413	68CC	5A			DECB	
1393	1414	68CD	26	F8		BNE	DCL0P2
1394	1415	68CF	35	90		PULS	X,PC
1395	1416						
1396	1417				*-----		

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1397 1418      *X:top of FILE NAME
1398 1419      *return C=1:match,C=0:not match
1399 1420 68D1      CMPFN    EQU      *
1400 1421 68D1 34 10      PSHS      X
1401 1422 68D3 108E 604F      LDY      #FNAME
1402 1423 68D7 C6 08      LDB      #8
1403 1424 68D9 A6 80  CMPLOP LDA      ,X+
1404 1425 68DB 81 E5      CMPA      #$E5      delete mark?
1405 1426 68DD 27 0F      BEQ      NOTM
1406 1427 68DF 81 20      CMPA      #'
1407 1428 68E1 27 07      BEQ      MATCH
1408 1429 68E3 A1 A0      CMPA      ,Y+
1409 1430 68E5 26 07      BNE      NOTM
1410 1431 68E7 5A      DECB
1411 1432 68E8 26 EF      BNE      CMPLOP
1412 1433 68EA 1A 01  MATCH ORCC      #$01      C=1
1413 1434 68EC 20 02      BRA      CMPEND
1414 1435 68EE 1C FE  NOTM ANDCC      #$FE      C=0
1415 1436 68F0 35 90  CMPEND PULS      X,PC
1416 1437
1417 1438      *-----
1418 1439 68F2      FNMIN    EQU      *
1419 1440      *clr buf
1420 1441 68F2 8E 605A      LDX      #FNAME+11
1421 1442 68F5 C6 0B      LDB      #8+3
1422 1443 68F7 86 20      LDA      #'
1423 1444 68F9 A7 82  FNIN0 STA      , -X
1424 1445 68FB 5A      DECB
1425 1446 68FC 26 FB      BNE      FNIN0
1426 1447      *fname in
1427 1448 68FE BD 690B  FNIN1 JSR      INOUTC
1428 1449 6901 A7 80      STA      ,X+
1429 1450 6903 81 0D      CMPA      #$0D
1430 1451 6905 27 03      BEQ      FNIEND
1431 1452 6907 5A      DECB
1432 1453 6908 26 F4      BNE      FNIN1
1433 1454 690A 39      FNIEND RTS
1434 1455
1435 1456 690B BD FBB8  INOUTC JSR      INCHA

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1436 1457 690E BD FBBB JSR OUTCHA
1437 1458 6911 39 RTS
1438 1459
1439 1460 *****
1440 1461 *===== parameter =====
1441 1462 6912 40 CMD0 FCB $40,$00,$00,$00,$00,$95 '@
1442 1463 6918 41 CMD1 FCB $41,$00,$00,$00,$00,$FF 'A
1443 1464 691E 51 CMD17 FCB $51,$00,$00,$00,$00,$FF 'Q
1444 1465 6924 58 CMD24 FCB $58,$00,$00,$00,$00,$FF 'X
1445 1466 692A 48 CMD8 FCB $48,$00,$00,$01,$AA,$87 'H
1446 1467 6930 7A CMD58 FCB $7A,$00,$00,$00,$00,$FF 'z
1447 1468 6936 77 CMD55 FCB $77,$00,$00,$00,$00,$FF 'w
1448 1469 693C 69 ACMD41 FCB $69,$40,$FF,$80,$00,$FF 'i
1449 1470 6942 49 CMD9 FCB $49,$00,$00,$00,$00,$FF 'I
1450 1471 *
1451 1472 6948 43 MCMD0 FCC /CMD0/, $D, $A
1452 1473 694E 04 FCB 4
1453 1474 694F 43 MCMD8 FCC /CMD8/, $D, $A
1454 1475 6955 04 FCB 4
1455 1476 6956 43 MCMD58 FCC /CMD58/, $D, $A
1456 1477 695D 04 FCB 4
1457 1478 695E 43 MCMD55 FCC /CMD55/, $D, $A
1458 1479 6965 04 FCB 4
1459 1480 6966 41 MCMD41 FCC /ACMD41/, $D, $A
1460 1481 696E 04 FCB 4
1461 1482 696F 43 MCMD9 FCC /CMD9/, $D, $A
1462 1483 6975 04 FCB 4
1463 1484 *
1464 1485 6976 0D MSMENU1 FCC $0D,$0A,/1:Read Sct, 2:Get SD
1465 data, 3:SD Init, 4:Disp MBR, 5:Disp BPB,/
1466 1486 69B5 04 FCB 4
1467 1487 69B6 0D MSMENU2 FCC $0D,$0A,/6:Disp FAT, 7:Disp DIR,
1468 8:Load FILE, 0:End : /
1469 1488 69E5 04 FCB 4
1470 1489 69E6 53 MSSCT FCC /SctNo: /
1471 1490 69ED 04 FCB 4
1472 1491 69EE 57 MSWROK FCC /Write OK?: /
1473 1492 69F9 04 FCB 4
1474 1493 69FA 0D MSSD0 FCC $0D,$0A,/SD Card/,$0D,$0A

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1475 1494 6A05 04          FCB      4
1476 1495 6A06 0D      MSSD1  FCC      $0D,$0A,/SDHC Card/,$0D,$0A
1477 1496 6A13 04          FCB      4
1478 1497 6A14 0D      MSINIT  FCC      $0D,$0A,$0D,$0A,/SD Card
1479      initializing.../
1480 1498 6A2F 04          FCB      4
1481 1499 6A30 53      MERR     FCC      /SD Initialize error!/, $0D,$0A
1482 1500 6A46 04          FCB      4
1483 1501 6A47 49      MSERR    FCC      /Input error!/, $0D,$0A
1484 1502 6A55 04          FCB      4
1485 1503 6A56 0D      MSOFS    FCC      $0D,$0A,/Sector no is:/
1486 1504 6A65 04          FCB      4
1487 1505 6A66 0D      MSBPB    FCC      $0D,$0A,/ BPB: /
1488 1506 6A6E 04          FCB      4
1489 1507 6A6F 0D      MSFAT    FCC      $0D,$0A,/ FAT: /
1490 1508 6A77 04          FCB      4
1491 1509 6A78 0D      MSDIR    FCC      $0D,$0A,/ DIR: /
1492 1510 6A80 04          FCB      4
1493 1511 6A81 0D      MSCL2    FCC      $0D,$0A,/ 1stCLUSTER: /
1494 1512 6A90 04          FCB      4
1495 1513 6A91 0D      MSINFN   FCC      $0D,$0A,$0D,$0A,/Input File Name:
1496 /
1497 1514 6AA6 04          FCB      4
1498 1515 6AA7 0D      MSLDED   FCC      $0D,$0A,/ Load end/
1499 1516 6AB2 04          FCB      4
1500 1517 6AB3 0D      MSNOFL   FCC      $0D,$0A,/ Not exist!/
1501 1518 6AC0 04          FCB      4
1502 1519 6AC1 0D      MSNXTTP  FCC      $0D,$0A,"Next page (SPC/else):
1503 "
1504 1520 6AD9 04          FCB      4
1505 1521
1506 1522      *****
1507 1523 6ADA FF      INPBUF  RMB      8
1508 1524
1509 1525      END      START
1510 0 error(s), 0 warning(s) is detected.
1511 Label address
1512 ACMD41 693C AD1      67B1 AD2      67C2 AD3      67CC AD4      67D2
1513 ADD0    65FB ADD1    6609 ADD2    6611 ADD3    6619 ADD40H 65F9

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1514	ADD42	67AB	ADX0	65EA	ADX1	65F0	ADX2	65F4	ADX3	65F8
1515	ASC411	6195	ASCMD41	6188	BASADRP	0033	BI21	6712	BI22	6718
1516	CALCH1	66E3	CALCHC	66DA	CALCSD	66CF	CL1	6758	CLEND	6768
1517	CLLOP	6763	CLOP	6736	CLSTH	6047	CLSTL	6049	CMD0	6912
1518	CMD1	6918	CMD17	691E	CMD24	6924	CMD55	6936	CMD58	6930
1519	CMD8	692A	CMD9	6942	CMPEND	68F0	CMPFN	68D1	CMPL0P	68D9
1520	CNT	604B	CNTSV	604C	CPY1	6627	CRLF	6727	CSR0N	FBF7
1521	CVBEND	67A5	CVBIN	6793	CVERR	67A8	CVT1	6771	DB2	6841
1522	DB2ASC	6832	DBLOP	6838	DCL0P1	68B8	DCL0P2	68C7	DCPY	64D1
1523	DIRENT	601E	DPGFLG	604E	DRET0	63B6	DRET1	647B	DSEND	68A5
1524	DSKP1	63D6	DSL0P1	687A	DSL0P2	688F	DSPB1	6663	DSPB2	6665
1525	DSPB3	6676	DSPBIN	66FE	DSPBPB	6386	DSPBUF	6659	DSPC1	6432
	DSPCKE	63EA	DSPCLS	68AC	DSPDIR	63AA	DSPEND	646B	DSPFAT	6398
	DSPFL0	63BD	DSPFL1	63D6	DSPFL2	63F4	DSPFL3	63F6	DSPFL4	6405
	DSPFL5	6452	DSPMBR	632C	DSPSIZ	684D	DSPYMD	67EC	DSS1	686A
	DSS12	6883	DSS13	6888	DSS14	688D	DSS2	6898	DSS3	689D
	DYMD1	67FE	DYMD2	680D	EEND	6792	EREND0	678B	EREND1	678D
	FILSIZ	6045	FNAME	604F	FNIEND	690A	FNIN0	68F9	FNIN1	68FE
	FNMIN	68F2	GET1	6337	GET2	6339	GET8CH	6732	GETBPB	636D
	GETD1	6697	GETDAT	6693	GETMBR	634A	GETSD	628B	GPL0P	6350
	GPL0P2	6378	GR00	6100	GR2580	61C2	GR2581	61D9	GR2582	61DB
	GR410	619D	GR550	617E	GR580	6154	GR581	615E	GR582	6160
	GR80	6124	GR81	6136	GR90	629E	GR91	62AD	IL0P1	61FE
	IL0P2	6211	IL0P3	622B	IL0P4	6269	INCHA	FBB8	INCX4	65EA
	INEND	6744	INIT	607E	INOUTC	690B	INPBUF	6ADA	INSCT	66AD
	ISTT	60E6	LD1	64B3	LD2	64CB	LD3	64F6	LD4	6501
	LD41	651F	LD5	6538	LD6	653E	LD7	6549	LD8	657E
	LD9	6586	LDE1	659B	LDEND	6595	LDFILE	646F	LDFL0	648B
	LDFL1	649F	LDFL5	64C0	LDNEND	659F	MAIN	605A	MATCH	68EA
	MCFLG	604D	MCMD0	6948	MCMD41	6966	MCMD55	695E	MCMD58	6956
	MCMD8	694F	MCMD9	696F	MEMSET	65A7	MEN1	609A	MEN2	60A2
	MEN3	60AA	MEN4	60B1	MEN5	60B9	MEN6	60C1	MEN7	60C9
	MEN8	60D1	MENU	6080	MERR	6A30	MS1	65B8	MS2	65C5
	MS3	65E4	MSBPB	6A66	MSCL2	6A81	MSDIR	6A78	MSERR	6A47
	MSFAT	6A6F	MSINFN	6A91	MSINIT	6A14	MSLDED	6AA7	MSLOP	65D7
	MSLOP0	65D3	MSMENU1	6976	MSMENU2	69B6	MSNOFL	6AB3	MSNXTTP	6AC1
	MSOFS	6A56	MSSCT	69E6	MSSD0	69FA	MSSD1	6A06	MSWROK	69EE
	NOTM	68EE	OKEND	6318	OUTCHA	FBBB	PBCTL	FD9F	PBDAT	FD9E
	PSEND	67E9	PSLOP	67DE	PSTRNG	67DC	RD1	6636	RD1SCT	661E

RD2	663D	RD3	664D	RESDAT	6007	RPT41	6169	RWBUF	6D00
SC01	60F7	SC2581	61BA	SC551	6176	SC581	614C	SC80	611C
SC91	6296	SCL0P1	62CD	SCL0P2	62E0	SCL0P3	62F3	SCL0P4	6306
SCMD0	60EF	SCMD258	61AD	SCMD55	6169	SCMD58	613F	SCMD8	610F
SCPY1	6251	SCTBPB	6009	SCTCLT	6019	SCTDIR	6011	SCTDSV	6015
SCTFAT	600D	SCTN0	6003	SCTN1	6004	SCTN2	6005	SCTN3	6006
SCTPCL	601D	SD1	61EF	SDERR	6312	SDHC	6008	SDINIT	60D9
SDRD	6319	SELOP	6362	SETCKH	66F5	SETCKL	66EC	SETSCT	66B5
SETSNO	635C	SFL0P	6751	SFT1	66C5	SKP1	623D	SKP2	6282
SNDD1	667E	SNDDAT	6677	START	6000	STTADR	6000	TXTADR	2000
VN	6002	WFSIZ	6041	WK1	6058	WMDY	603E		

<!--temp--><!--><!--><!--><!-->

SDHC9FM

As I mentioned in my previous report, "An Attempt to Connect an RTC and SD Card to the FM-7 (RTC Edition) (March 9, 2023)," the fact that you can use an RTC and SD card simply by adding one 6821PIA to the FM-7 is quite a bargain, wouldn't you say?